THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET A

Chapter 10: **Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. On a map, 1 cm represents 25 km. A distance of 4 cm on the map represents: 1

(a) 200 km **(b)** 50 km **(c)** 125 km **(d)** 100 km

2. On a map, 1 cm represents 50 km. A distance of 2.4 cm on the map represents: 1

(a) 60 km **(b)** 170 km **(c)** 240 km **(d)** 120 km

3. ₹ 900 is shared among A, B and C in the ratio 3:4:2. B gets: 1

(a) ₹ 200 **(b)** ₹ 300 **(c)** ₹ 600 **(d)** ₹ 400

4. 12 workers finish a job in 10 days. How many days will 8 workers take? 1

(a) 15 **(b)** 17 **(c)** 14 **(d)** 6.66667

5. 20 workers finish a job in 6 days. How many days will 8 workers take? 1

(a) 17 **(b)** 14 **(c)** 2.4 **(d)** 15

6. In a pie chart, a sector representing 40% of the data has a central angle of: 1

(a) 154° **(b)** 80° **(c)** 144° **(d)** 40°

7. If two quantities are inversely proportional, their: 1

(a) product is constant **(b)** sum is constant
(c) difference is constant **(d)** ratio is constant

8. The ratio 20 : 30 : 50 in simplest form is: 1

(a) 2 : 3 : 4 **(b)** 4 : 6 : 10 **(c)** 2 : 3 : 5 **(d)** 1 : 2 : 3

9. Which pair is in direct proportion? 1

(a) Pipes and time **(b)** Cost and number of identical items
(c) Workers and days **(d)** Speed and time

10. The map distance is 5 cm and the actual distance is 20 km. The scale is 1 cm = 1

(a) 5 km **(b)** 100 km **(c)** 0.25 km **(d)** 4 km

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is an inverse proportion? Give an example. 2

12. A map has scale 1 cm = 8 km. Find the actual distance for 12.5 cm. 2

13. Divide 360 in the ratio 1 : 2 : 3. 2

14. Write the ratio 2 : 3 and 3 : 5 as a single three-term ratio. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Divide ₹ 1800 among A, B and C in the ratio 2:3:4. 3

16. Divide ₹ 4400 among A, B and C in the ratio 1:3:7. 3

17. Divide ₹ 2700 among A, B and C in the ratio 4:5:9. 3

OR

Divide ₹ 3600 among A, B and C in the ratio 2:5:5. 3

18. Divide ₹ 3000 among A, B and C in the ratio 1:2:7. 3

OR

Divide ₹ 4800 among A, B and C in the ratio 3:4:5. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Explain inverse proportion with an example. 5
(ii) 8 men can dig a trench in 15 days. In how many days can 12 men dig it?
(iii) How many men are needed to finish in 6 days?

OR

Case study: A road map uses a scale of 1 : 500000. 5
(i) Express in cm to km.
(ii) Two cities are 18 cm apart; find the actual distance.
(iii) A bus at 50 km/h takes how long?

20. (i) Divide ₹ 12000 among A, B and C so that A : B = 2 : 3 and B : C = 4 : 5. 5
(ii) Who gets the largest share?

OR

(i) How is a pie chart drawn? 5
(ii) Draw a pie chart for expenses of ₹ 2400: food 1200, rent 600, travel 400, savings 200 and give angles.

END OF QUESTION PAPER — Total: 40 Marks

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